

1 **Random generalized Lotka-Volterra model on sparse graphs:**
2 **Gaussian expansion of dynamic cavity method**

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Abstract

An article usually includes an abstract, a concise summary of the work covered at length in the main body of the article.

I. INTRODUCTION

Intro on the subject.

II. METHOD

Explain here the small-coupling method and population dynamics.

III. RESULTS

Present here results

A. Random generalized Lotka-Volterra model on sparse graphs

We consider a generalized Lotka-Volterra (gLV) model describing the dynamics of an interacting community of N species, labeled by $i = 1, \dots, N$. The species interacts on an undirected random graph G with symmetric adjacency matrix $A = \{a_{ij}\}$, meaning that species i and j are interacting if $a_{ij} = a_{ji} = 1$, otherwise the matrix elements are zero. The time-dependent number density of individuals of species i is denoted by $x_i(t)$, and evolves in time according to

$$\frac{d}{dt}x_i(t) = x_i(t) \left[1 - x_i(t) + \sum_{j \neq i} a_{ij} J_{ij} x_j(t) \right] + \eta_i(t) + \lambda, \quad (1)$$

where we included, for the sake of generality, a Gaussian noise $\eta_i(t)$ with vanishing mean and variance $\langle \eta_i(t) \eta_{i'}(t') \rangle = 2Tg(x_i(t))\delta_{ii'}\delta(t - t')$, and a constant immigration rate λ . We will consider in what follows three choices for the function $g(x)$: (i) $g(x) = 0$ for deterministic gLV model, (ii) $g(x) = x$ for demographic noise, and (iii) $g(x) = x^2$ for environmental noise.

The couplings J_{ij} represent the (per capita) effect of species j on one another. A negative coefficient J_{ij} indicates a competitive effect of species j on species i .

We consider here Gaussian J_{ij} with average $\overline{J_{ij}} = \overline{J_{ji}} = m/K$, variance $\text{Var}(J_{ij}) = \text{Var}(J_{ji}) = \sigma^2/K$ and covariance $\text{Corr}(J_{ij}, J_{ji}) = \gamma$. The overbar $\overline{}$ represents the average over the random couplings ensemble. The scaling of the moments with the average degree of the graph K is needed to produce a well-defined fully connected limit $K \rightarrow \infty$. The parameter $-1 \leq \gamma \leq 1$, being related to the correlation between the interaction coefficients J_{ij} and J_{ji} , measures the fraction of predator-prey pairs. Such a predator-prey pair consists of two species i and j for which J_{ij} and J_{ji} have opposite signs. If for example J_{ij} is positive and J_{ji} is negative, the presence of species i has a detrimental effect on species j , while the presence of species j is beneficial for species i . For any pair $i < j$ we can write

$$J_{ij} = \frac{m}{K} + \frac{\sigma}{\sqrt{K}} z_{ij}, \quad J_{ji} = \frac{m}{K} + \frac{\sigma}{\sqrt{K}} z_{ji}, \quad (2)$$

where z_{ij} and z_{ji} are Gaussian random variables with $\overline{z_{ij}} = 0$, $\overline{z_{ij}^2} = 1$, $\overline{z_{ij} z_{ji}} = \gamma$.

For $\gamma = 1$ we have $z_{ij} = z_{ji}$ with probability one, for $\gamma = 0$ z_{ij} and z_{ji} are uncorrelated, and for $\gamma = -1$ we have $z_{ij} = -z_{ji}$ with probability one. The latter, $\gamma = -1$, corresponds to having all the interactions of the predator-prey type. Increasing the value of γ this fraction decrease, until the opposite case, $\gamma = 1$ of no predator-prey type interactions. The other interactions can be of a mutualistic type (J_{ij} and J_{ji} both positive) or strictly competitive (J_{ij} and J_{ji} both negative).

B. Small coupling expansion of dynamic cavity equations

We use a small coupling expansion of the dynamic cavity equations in a way similar to [1] to derive an effective Langevin equation for non-interactive species. This approach begins by constructing the Martin-Siggia-Rose-Janssen-De Dominicis (MSRJD) dynamic partition function through path integrals of the trajectories [2–4]

$$Z = \left\langle \int \mathcal{D}\vec{x} \prod_i p_i(x_i(0)) \delta \left(\dot{x}_i(t) - x_i(t) \left[1 - x_i(t) + \sum_{j \neq i} a_{ij} J_{ij} x_j(t) + h_i(t) \right] - \eta_i(t) - \lambda \right) \right\rangle_{\vec{\eta}}, \quad (3)$$

where $\int \mathcal{D}\vec{x} = \prod_i \int \mathcal{D}x_i$ is a functional integral over the trajectories of the system, and $h_i(t)$ is an external field, needed to compute response functions. Following the steps of [1] we put forward a dynamic cavity Ansatz and expand the cavity marginals for small couplings. This results in a set of effective de-coupled Langevin equations on the cavity graph. For any edge (i, j) on the graph G the effective cavity population size $x_{i \setminus j}(t)$ evolves under the following stochastic process

$$\begin{aligned} \frac{d}{dt} x_{i \setminus j}(t) = & x_{i \setminus j}(t) \left[1 - x_{i \setminus j}(t) + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}(t) + \sum_{k \in \partial i \setminus j} \int_0^t dt' J_{ik} J_{ki} R_{k \setminus i}(t, t') x_{i \setminus j}(t') + h_{i \setminus j}(t) + \zeta_{i \setminus j}(t) \right] \\ & + \sqrt{2Tg(x_{i \setminus j}(t))} \xi_{i \setminus j}(t) + \lambda \end{aligned} \quad (4)$$

where $h_{i \setminus j}(t)$ is an external field, needed to compute response functions, and $\zeta_{i \setminus j}$ and $\xi_{i \setminus j}$ are Gaussian noises with vanishing mean and correlations

$$\langle \zeta_{i \setminus j}(t) \zeta_{i \setminus j}(t') \rangle_{i \setminus j} = \sum_{k \in \partial i \setminus j} J_{ik}^2 C_{k \setminus i}(t, t'), \quad \langle \xi_{i \setminus j}(t) \xi_{i \setminus j}(t') \rangle_{i \setminus j} = \delta(t - t'). \quad (5)$$

The terms $\mu_{k \setminus i}(t)$, $R_{k \setminus i}(t, t')$, and $C_{k \setminus i}(t, t')$ are respectively the cavity mean abundance, the cavity response function, and the cavity (connected) two-point correlation function. They are obtained self-consistently from the effective dynamics of the neighbors

$$\mu_{k \setminus i}(t) = \langle x_{k \setminus i}(t) \rangle_{k \setminus i}, \quad R_{k \setminus i}(t, t') = \frac{\delta}{\delta h_{k \setminus i}(t')} \langle x_{k \setminus i}(t) \rangle_{k \setminus i}, \quad C(t, t') = \langle x_{k \setminus i}(t) x_{k \setminus i}(t') \rangle_{k \setminus i} - \mu_{k \setminus i}(t) \mu_{k \setminus i}(t'), \quad (6)$$

where $\langle \dots \rangle_{k \setminus i}$ describes the average over the distribution of the effective process Eq. (4) for the edge (k, i) . The effective process is therefore de-coupled, and the interactions are captured by the self-consistent external fields $\mu_{k \setminus i}(t)$, $R_{k \setminus i}(t, t')$, and $C_{k \setminus i}(t, t')$. A detailed derivation can be found in Appendix A.

Once the cavity averages have been determined through self-consistency, one can obtain the effective de-coupled Langevin equations for the species population

$$\frac{d}{dt} x_i(t) = x_i(t) \left[1 - x_i(t) + \sum_{k \in \partial i} J_{ik} \mu_{k \setminus i}(t) + \sum_{k \in \partial i} \int_0^t dt' J_{ik} J_{ki} R_{k \setminus i}(t, t') x_i(t') + h_i(t) + \zeta_i(t) \right] + \sqrt{2Tg(x_i(t))} \xi_i(t) + \lambda, \quad (7)$$

with Gaussian noises with vanishing mean and correlations

$$\langle \zeta_i(t) \zeta_i(t') \rangle_i = \sum_{k \in \partial i} J_{ik}^2 C_{k \setminus i}(t, t'), \quad \langle \xi_i(t) \xi_i(t') \rangle_i = \delta(t - t'). \quad (8)$$

The average $\langle \dots \rangle_i$ is performed over the effective de-coupled process Eq. (7).

C. Numerical integration

We propose a numerical integration scheme, mostly based on what has been proposed in [5]. The algorithm iteratively updates the self-consistent external fields until convergence.

We start with initial guesses for the cavity mean abundances $\{\mu_{i\setminus j}\}_{(i,j)\in E}$, the cavity correlations $\{C_{i\setminus j}\}_{(i,j)\in E}$, and the cavity responses $\{R_{i\setminus j}\}_{(i,j)\in E}$ for each edge of the graph. At iteration n , for each edge (i, j) the following steps are followed:

1. We sample N_{traj} Gaussian paths $(\zeta_{i\setminus j}(t), \xi_{i\setminus j}(t))$, where $\zeta_{i\setminus j}(t)$ and $\xi_{i\setminus j}(t)$ have, respectively, covariance matrices $C_{i\setminus j}^\zeta(t, t') = \sum_{k \in \partial i \setminus j} J_{ik}^2 C_{k\setminus i}(t, t')$ and $C_{i\setminus j}^\xi(t, t') = \delta(t, t')$.
2. For each path, we numerically integrate the effective cavity equation with an initial condition sampled according to $p_0(x)$:

$$\begin{aligned} \frac{d}{dt} x_{i\setminus j}(t) &= x_{i\setminus j}(t) \left[1 - x_{i\setminus j}(t) + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k\setminus i}(t) + \sum_{k \in \partial i \setminus j} \int_0^t dt' J_{ik} J_{ki} R_{k\setminus i}(t, t') x_{i\setminus j}(t') + \zeta_{i\setminus j}(t) \right] \\ &\quad + \sqrt{2Tg(x_{i\setminus j}(t))} \xi_{i\setminus j}(t) + \lambda, \\ x_{i\setminus j}(0) &\sim p_0(x_{i\setminus j}(0)). \end{aligned}$$

3. We compute the new values of the cavity self-consistent fields as averages over the paths:

$$\begin{aligned} \mu_{i\setminus j}^{\text{new}}(t) &= \mathbb{E}_{\text{paths}} [x_{i\setminus j}(t)], \\ C_{i\setminus j}^{\text{new}}(t, t') &= \mathbb{E}_{\text{paths}} [x_{i\setminus j}(t) x_{i\setminus j}(t')] - \mu_{i\setminus j}^{\text{new}}(t) \mu_{i\setminus j}^{\text{new}}(t'), \\ R_{i\setminus j}^{\text{new}}(t, t') &= \mathbb{E}_{\text{paths}} [R_{i\setminus j}^{\text{sp}}(t, t')]. \end{aligned}$$

4. We update softly the cavity self-consistent fields: $X_{i\setminus j}^{\text{updated}} = (1 - d)X_{i\setminus j}^{\text{new}} + dX_{i\setminus j}$, where X is μ , C and R . d is the damping factor.

The terms $R_{i\setminus j}^{\text{sp}}(t, t')$ are single-path response functions, and can be computed through numerical integration of the equation obtained by directly applying $\delta/\delta h_{i\setminus j}(t')$ to Eq. (4). More details can be found in Appendix B.

The algorithm proceeds in iterations, where each iteration consists of updating all edges (i, j) of the graph. These iterations repeat until convergence is achieved, defined as the change in the cavity self-consistent fields between successive iterations dropping below a threshold ε . For computational efficiency, we evaluate convergence based on the stabilization of correlation functions only.

$$\max_{(i,j)} \|C_{i\setminus j}^{\text{updated}} - C_{i\setminus j}^{\text{new}}\| < \epsilon, \quad (9)$$

where $\|\cdot\|$ is some norm to be defined. When convergence is reached, we can compute the local effective process by using the self-consistent cavity fields in Eq. (7). In practice, for each node i of the graph we sample N_{traj} Gaussian paths $(\zeta_i(t), \xi_i(t))$, where $\zeta_i(t)$ and $\xi_i(t)$ have, respectively, covariance matrices $C_i^\zeta(t, t') = \sum_{k \in \partial i} J_{ik}^2 C_{k\setminus i}(t, t')$ and $C_i^\xi(t, t') = \delta(t, t')$. We then numerically integrate the effective marginal equation with an initial condition sampled according to $p_0(x)$:

$$\begin{aligned} \frac{d}{dt} x_i(t) &= x_i(t) \left[1 - x_i(t) + \sum_{k \in \partial i} J_{ik} \mu_{k\setminus i}(t) + \sum_{k \in \partial i} \int_0^t dt' J_{ik} J_{ki} R_{k\setminus i}(t, t') x_i(t') + \zeta_i(t) \right] \\ &\quad + \sqrt{2Tg(x_i(t))} \xi_i(t) + \lambda, \\ x_i(0) &\sim p_0(x_i(0)). \end{aligned}$$

The full average quantities are then obtained by averaging over the sampled paths:

$$\begin{aligned} \mu_i(t) &= \mathbb{E}_{\text{paths}} [x_i(t)], \\ C_i(t, t') &= \mathbb{E}_{\text{paths}} [x_i(t) x_i(t')] - \mu_i(t) \mu_i(t'), \\ R_i(t, t') &= \mathbb{E}_{\text{paths}} [R_i^{\text{sp}}(t, t')]. \end{aligned}$$

More details on the numerical integration can be found in Appendix C.

D. Gaussian closure approximation

While a direct numerical integration of the effective process is possible, we can also use a Gaussian closure approximation to obtain analytical results. This approach is based on the assumption that the distribution of the population sizes $x_i(t)$ is Gaussian, which allows us to compute the moments of the process analytically. The Gaussian closure approximation leads to a set of self-consistent equations for the first and second moments of the population sizes, which can be solved iteratively.

In particular, by taking averages of the effective cavity process Eq. (4) and approximating three-times averages as sums of products of one- and two-times averages, we obtain the following self-consistent equations for the first and second moments of the population sizes:

$$\begin{aligned} \frac{d}{dt}\mu_{i\setminus j}(t) &= \langle \dot{x}_{i\setminus j}(t) \rangle_{i\setminus j} = \mu_{i\setminus j}(t) - C_{i\setminus j}(t, t) - \mu_{i\setminus j}^2(t) + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k\setminus i}(t) \mu_{i\setminus j}(t) \\ &+ \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^t ds R_{k\setminus i}(t, s) (C_{i\setminus j}(t, s) + \mu_{i\setminus j}(t) \mu_{i\setminus j}(s)) + \lambda, \end{aligned} \quad (10)$$

$$\begin{aligned} \frac{\partial}{\partial t} C_{i\setminus j}(t, t') &= \langle \dot{x}_{i\setminus j}(t) x_{i\setminus j}(t') \rangle_{i\setminus j} = (1 - 2\mu_{i\setminus j}(t)) C_{i\setminus j}(t, t') + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k\setminus i}(t) C_{i\setminus j}(t, t') \\ &+ \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^t ds R_{k\setminus i}(t, s) (\mu_{i\setminus j}(t) C_{i\setminus j}(s, t') + C_{i\setminus j}(t, t') \mu_{i\setminus j}(s)) + 2T \mathcal{L}_{i\setminus j}(t) R_{i\setminus j}(t', t) \\ &+ \sum_{k \in \partial i \setminus j} J_{ik}^2 \left(\int_0^{t'} ds C_{k\setminus i}(t, s) \mu_{i\setminus j}(t) R_{i\setminus j}(t', s) + \int_0^t ds C_{k\setminus i}(t, s) R_{i\setminus j}(t, s) \mu_{i\setminus j}(t') \right), \end{aligned} \quad (11)$$

$$\begin{aligned} \frac{\partial}{\partial t} R_{i\setminus j}(t, t') &= \langle \dot{x}_{i\setminus j}(t) \dot{x}_{i\setminus j}(t') \rangle_{i\setminus j} = (1 - 2\mu_{i\setminus j}(t)) R_{i\setminus j}(t, t') + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k\setminus i}(t) R_{i\setminus j}(t, t') + \delta(t - t') \\ &+ \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \left(\int_0^t ds R_{k\setminus i}(t, s) R_{i\setminus j}(t, t') \mu_{i\setminus j}(s) + \int_{t'}^t ds \mu_{i\setminus j}(t) R_{k\setminus i}(t, s) R_{i\setminus j}(s, t') \right), \end{aligned} \quad (12)$$

where we have introduced the function

$$\mathcal{L}_{i\setminus j}(t) = \begin{cases} 0 & g(x) = 0 & \text{deterministic} \\ 1 & g(x) = x & \text{demographic} \\ \mu_{i\setminus j}(t) & g(x) = x^2 & \text{environmental.} \end{cases} \quad (13)$$

An equivalent set of equations can be obtained for the full marginal process:

$$\begin{aligned} \frac{d}{dt}\mu_i(t) &= \langle \dot{x}_i(t) \rangle_i = \mu_i(t) - C_i(t, t) - \mu_i^2(t) + \sum_{k \in \partial i} J_{ik} \mu_{k\setminus i}(t) \mu_i(t) \\ &+ \sum_{k \in \partial i} J_{ik} J_{ki} \int_0^t ds R_{k\setminus i}(t, s) (C_i(t, s) + \mu_i(t) \mu_i(s)) + \lambda, \end{aligned} \quad (14)$$

$$\begin{aligned} \frac{\partial}{\partial t} C_i(t, t') &= \langle \dot{x}_i(t) x_i(t') \rangle_i = (1 - 2\mu_i(t)) C_i(t, t') + \sum_{k \in \partial i} J_{ik} \mu_{k\setminus i}(t) C_i(t, t') \\ &+ \sum_{k \in \partial i} J_{ik} J_{ki} \int_0^t ds R_{k\setminus i}(t, s) (\mu_i(t) C_i(s, t') + C_i(t, t') \mu_i(s)) + 2T \mathcal{L}_i(t) R_i(t', t) \\ &+ \sum_{k \in \partial i} J_{ik}^2 \left(\int_0^{t'} ds C_{k\setminus i}(t, s) \mu_i(t) R_i(t', s) + \int_0^t ds C_{k\setminus i}(t, s) R_i(t, s) \mu_i(t') \right), \end{aligned} \quad (15)$$

$$\begin{aligned} \frac{\partial}{\partial t} R_i(t, t') &= \langle \dot{x}_i(t) \dot{x}_i(t') \rangle_i = (1 - 2\mu_i(t)) R_i(t, t') + \sum_{k \in \partial i} J_{ik} \mu_{k\setminus i}(t) R_i(t, t') + \delta(t - t') \\ &+ \sum_{k \in \partial i} J_{ik} J_{ki} \left(\int_0^t ds R_{k\setminus i}(t, s) R_i(t, t') \mu_i(s) + \int_{t'}^t ds \mu_i(t) R_{k\setminus i}(t, s) R_i(s, t') \right), \end{aligned} \quad (16)$$

where the function $\mathcal{L}_i(t)$ is defined as

$$\mathcal{L}_i(t) = \begin{cases} 0 & g(x) = 0 & \text{deterministic} \\ 1 & g(x) = x & \text{demographic} \\ \mu_i(t) & g(x) = x^2 & \text{environmental.} \end{cases} \quad (17)$$

1. Stationary state

Assuming the system reaches a stationary state, we can set $\mu(t) = \mu$, $C(t, t') = C_i(t - t')$, and $R(t, t') = R(t - t')$. This corresponds to assuming the system to be in the single equilibrium state. The self-consistent cavity equations then simplify to

$$\begin{aligned} \frac{d}{dt}\mu_{i \setminus j} = 0 &= \mu_{i \setminus j} \left(1 - \mu_{i \setminus j} + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i} + \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \chi_{k \setminus i} \mu_{i \setminus j} \right) - C_{i \setminus j}(0) \\ &+ \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^\infty du R_{k \setminus i}(u) C_{i \setminus j}(u) + \lambda, \end{aligned} \quad (18)$$

$$\begin{aligned} \frac{d}{d\tau} C_{i \setminus j}(\tau) &= C_{i \setminus j}(\tau) \left(1 - 2\mu_{i \setminus j} + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i} + \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \chi_{k \setminus i} \mu_{i \setminus j} \right) + 2T \mathcal{L}_{i \setminus j} R_{i \setminus j}(-\tau) \\ &+ \mu_{i \setminus j} \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^\infty du R_{k \setminus i}(u) C_{i \setminus j}(\tau - u) \\ &+ \mu_{i \setminus j} \sum_{k \in \partial i \setminus j} J_{ik}^2 \int_0^\infty du (C_{k \setminus i}(u - \tau) R_{i \setminus j}(u) + C_{k \setminus i}(u) R_{i \setminus j}(u)), \end{aligned} \quad (19)$$

$$\begin{aligned} \frac{d}{d\tau} R_{i \setminus j}(\tau) &= R_{i \setminus j}(\tau) \left(1 - 2\mu_{i \setminus j} + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i} + \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \chi_{k \setminus i} \mu_{i \setminus j} \right) + \delta(\tau) \\ &+ \mu_{i \setminus j} \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^\tau du R_{k \setminus i}(u) R_{i \setminus j}(\tau - u), \end{aligned} \quad (20)$$

where we have introduced the time shift $\tau = t - t'$, and the susceptibility $\chi_{i \setminus j} = \int_0^\infty du R_{i \setminus j}(u)$, which is the integrated response of the population i in the cavity graph $\partial i \setminus j$ to a local perturbation. We have defined the function $\mathcal{L}_{i \setminus j}$ as

$$\mathcal{L}_{i \setminus j} = \begin{cases} 0 & g(x) = 0 & \text{deterministic} \\ 1 & g(x) = x & \text{demographic} \\ \mu_{i \setminus j} & g(x) = x^2 & \text{environmental.} \end{cases} \quad (21)$$

2. Fixed Point + Laplace

I have the intuition that we have some chances of going ahead exploiting Laplace as in the linear case for C and R , but having a fixed point in μ .

Assuming that μ reached the fixed point it is easy to show that:

$$\begin{aligned} p \hat{R}_{i \setminus j}(p) + \hat{R}_{i \setminus j}(0) &= \left[1 - 2\mu_{i \setminus j} + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i} \right] \hat{R}_{i \setminus j}(p) + \\ &+ \mu_{i \setminus j} \hat{R}_{i \setminus j}(p) \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} (\chi_{k \setminus i} + \hat{R}_{k \setminus i}(p)) \end{aligned} \quad (22)$$

Similarly:

$$\begin{aligned}
p\hat{C}_{i\setminus j}(p) + \hat{C}_{i\setminus j}(0) &= \left[1 - 2\mu_{i\setminus j} + \sum_{k \in \partial i \setminus j} J_{ik}\mu_{k\setminus i}\right] \hat{C}_{i\setminus j}(p) + \\
&+ \mu_{i\setminus j} \hat{C}_{i\setminus j}(p) \sum_{k \in \partial i \setminus j} J_{ik}J_{ki}(\chi_{k\setminus i} + \hat{R}_{k\setminus i}(p)) + \\
&+ \mu_{i\setminus j} \hat{R}_{i\setminus j}(p) \sum_{k \in \partial i \setminus j} J_{ik}^2 \hat{C}_{k\setminus i}(p) + \mu_{i\setminus j} \chi_{i\setminus j} \sum_{k \in \partial i \setminus j} J_{ik}^2 \hat{C}_{k\setminus i}(p)
\end{aligned} \tag{23}$$

114 where the last term in this expression is *dirty*. I assumed that: $\mathcal{L}\left[\int_0^t du C_{k\setminus i}(u) R_{k\setminus i}(u)\right] = \chi_{i\setminus j} \hat{C}_{k\setminus i}(p)$.

115 Finally, for the fixed point equation:

$$0 = \mu_{i\setminus j}(1 + \sum_{k \in \partial i \setminus j} J_{ik}\mu_{k\setminus i}) - \mu_{i\setminus j}^2(1 - \sum_{k \in \partial i \setminus j} J_{ik}J_{ki}\chi_{k\setminus i}) - q_{i\setminus j}(1 - \sum_{k \in \partial i \setminus j} J_{ik}J_{ki}\chi_{k\setminus i}) \tag{24}$$

116 that looks prettier this way:

$$0 = \mu_{i\setminus j}^2 - \mu_{i\setminus j} \frac{1 + \sum_{k \in \partial i \setminus j} J_{ik}\mu_{k\setminus i}}{1 - \sum_{k \in \partial i \setminus j} J_{ik}J_{ki}\chi_{k\setminus i}} + q_{i\setminus j} \tag{25}$$

117 One can try a bit more and look for fixed point equations also for C . I get for q :

$$0 = \left[1 - 2\mu_{i\setminus j} + \sum_{k \in \partial i \setminus j} J_{ik}\mu_{k\setminus i} + 2\mu_{i\setminus j} \sum_{k \in \partial i \setminus j} J_{ik}J_{ki}\chi_{k\setminus i}\right] q_{i\setminus j} + \mu_{i\setminus j} \chi_{i\setminus j} \sum_{k \in \partial i \setminus j} J_{ik}^2 q_{k\setminus i} \tag{26}$$

118 E. Fixed-point equations for the deterministic model

119 We focus on the deterministic model without immigration, where $g(x) = 0$ and $\lambda = 0$. In this case, for certain
120 values of the parameters m , σ and γ , we can assume that the system reaches a steady state which does not depend on
121 the initial condition. We further assume that the dynamics reaches a unique fixed-point $\lim_{t \rightarrow \infty} x_{i\setminus j}(t) = x_{i\setminus j}^*$, and
122 therefore $\lim_{t \rightarrow \infty} \zeta_{i\setminus j}(t) = \zeta_{i\setminus j}^*$. The fixed point values $x_{i\setminus j}^*$, $\zeta_{i\setminus j}^*$ are both random variables. It follows that the mean
123 cavity abundance and the cavity correlation become time-independent variables $\lim_{t \rightarrow \infty} \mu_{i\setminus j}(t) = \mu_{i\setminus j}^* = \langle x_{i\setminus j}^* \rangle_{i\setminus j}^*$,
124 and $\lim_{t, t' \rightarrow \infty} C_{i\setminus j}(t, t') = q_{i\setminus j}^* = \langle (x_{i\setminus j}^*)^2 \rangle_{i\setminus j}^* - (\mu_{i\setminus j}^*)^2$. Within this assumption, the cavity response function is time-
125 translational-invariant, i.e., $R_{i\setminus j}(t, t')$ depends only on the time difference $\tau = t - t'$. Due to causality, $R_{i\setminus j}(\tau) = 0$
126 for $\tau \leq 0$, and we define the susceptibility as $\chi_{i\setminus j}^* = \int_0^\infty d\tau R_{i\setminus j}^*(\tau)$.

127 The fixed point satisfies the self-consistent equation

$$x_{i\setminus j}^* \left[1 - x_{i\setminus j}^* + \sum_{k \in \partial i \setminus j} J_{ik}\mu_{k\setminus i}^* + \sum_{k \in \partial i \setminus j} J_{ik}J_{ki}\chi_{k\setminus i}^* x_{i\setminus j}^* + \xi_{i\setminus j}^* + h_{i\setminus j} \right] = 0, \tag{27}$$

128 where $h_{i\setminus j}$ is a constant external field needed to obtain the integrated cavity response, and the noise is Gaussian with
129 zero mean $\langle \xi_{i\setminus j}^* \rangle$ and variance $\langle (\xi_{i\setminus j}^*)^2 \rangle = \sum_{k \in \partial i \setminus j} J_{ik}^2 q_{k\setminus i}^* = \Sigma_{i\setminus j}^2$. We can therefore write $\xi_{i\setminus j}^* = -\Sigma_{i\setminus j} s$, where s is a
130 standard Gaussian variable with zero mean and unit variance. Eq. (27) admits non-vanishing solution

$$x_{i\setminus j}^*(s) = \frac{\Sigma_{i\setminus j}(\Delta_{i\setminus j} - s) + h_{i\setminus j}}{\Gamma_{i\setminus j}} \Theta\left(\frac{\Sigma_{i\setminus j}(\Delta_{i\setminus j} - s) + h_{i\setminus j}}{\Gamma_{i\setminus j}}\right), \tag{28}$$

131 where $\Theta(x)$ is the Heaviside step function, which ensures positivity of the fixed-point cavity abundance. We have
132 defined $\Delta_{i\setminus j} = (1 + \sum_{k \in \partial i \setminus j} J_{ik}\mu_{k\setminus i}^*)/\Sigma_{i\setminus j}$ and $\Gamma_{i\setminus j} = 1 - \sum_{k \in \partial i \setminus j} J_{ik}J_{ki}\chi_{k\setminus i}^*$.

By averaging over the Gaussian distribution of s we obtain a set of self-consistent equations for the parameters

$$\mu_{i \setminus j}^* = \frac{\Sigma_{i \setminus j}}{\Gamma_{i \setminus j}} \left[\Theta(\Gamma_{i \setminus j}) \int_{-\infty}^{\Delta_{i \setminus j}} Ds (\Delta_{i \setminus j} - s) + \Theta(-\Gamma_{i \setminus j}) \int_{\Delta_{i \setminus j}}^{\infty} Ds (\Delta_{i \setminus j} - s) \right] \quad (29a)$$

$$q_{i \setminus j}^* = \frac{\Sigma_{i \setminus j}^2}{\Gamma_{i \setminus j}^2} \left[\Theta(\Gamma_{i \setminus j}) \int_{-\infty}^{\Delta_{i \setminus j}} Ds (\Delta_{i \setminus j} - s)^2 + \Theta(-\Gamma_{i \setminus j}) \int_{\Delta_{i \setminus j}}^{\infty} Ds (\Delta_{i \setminus j} - s)^2 \right] - \left(\mu_{i \setminus j}^* \right)^2 \quad (29b)$$

$$\chi_{i \setminus j}^* = \frac{1}{\Gamma_{i \setminus j}} \left[\Theta(\Gamma_{i \setminus j}) \int_{-\infty}^{\Delta_{i \setminus j}} Ds + \Theta(-\Gamma_{i \setminus j}) \int_{\Delta_{i \setminus j}}^{\infty} Ds \right]. \quad (29c)$$

We observe that the integrals can be written in terms of the error function $\text{erf}(x) = \int_0^x ds e^{-s^2} 2/\sqrt{\pi}$. Defining $\varphi(x) = e^{-x^2/2}/\sqrt{2\pi}$ the normal distribution function and $\Phi(x) = [1 + \text{erf}(x/\sqrt{2})]/2$ its cumulative distribution function we obtain

$$\mu_{i \setminus j}^* = \frac{\Sigma_{i \setminus j}}{\Gamma_{i \setminus j}} \left[\Theta(\Gamma_{i \setminus j}) (\Delta_{i \setminus j} \Phi(\Delta_{i \setminus j}) + \varphi(\Delta_{i \setminus j})) + \Theta(-\Gamma_{i \setminus j}) (\Delta_{i \setminus j} \Phi(-\Delta_{i \setminus j}) - \varphi(-\Delta_{i \setminus j})) \right] \quad (30a)$$

$$q_{i \setminus j}^* = \frac{\Sigma_{i \setminus j}^2}{\Gamma_{i \setminus j}^2} \left[\Theta(\Gamma_{i \setminus j}) \left((1 + \Delta_{i \setminus j}^2) \Phi(\Delta_{i \setminus j}) + \Delta_{i \setminus j} \varphi(\Delta_{i \setminus j}) \right) + \Theta(-\Gamma_{i \setminus j}) \left((1 + \Delta_{i \setminus j}^2) \Phi(-\Delta_{i \setminus j}) - \Delta_{i \setminus j} \varphi(-\Delta_{i \setminus j}) \right) \right] - \left(\mu_{i \setminus j}^* \right)^2 \quad (30b)$$

$$\chi_{i \setminus j}^* = \frac{1}{\Gamma_{i \setminus j}} \left[\Theta(\Gamma_{i \setminus j}) \Phi(\Delta_{i \setminus j}) + \Theta(-\Gamma_{i \setminus j}) \Phi(-\Delta_{i \setminus j}) \right]. \quad (30c)$$

We can solve these equations by means of a message-passing algorithm, together with the definitions of $\Delta_{i \setminus j}$, $\Sigma_{i \setminus j}$ and $\Gamma_{i \setminus j}$, which are given by

$$\Delta_{i \setminus j} = \frac{1 + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}^*}{\sqrt{\sum_{k \in \partial i \setminus j} J_{ik}^2 q_{k \setminus i}^*}} \quad (31a)$$

$$\Sigma_{i \setminus j}^2 = \sum_{k \in \partial i \setminus j} J_{ik}^2 q_{k \setminus i}^* \quad (31b)$$

$$\Gamma_{i \setminus j} = 1 - \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \chi_{k \setminus i}^*. \quad (31c)$$

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Appendix A: Derivation of the effective cavity process

The calculation is based on the principles of [1]. The dynamical partition function of the Lotka-Volterra system can be written as

$$Z = \left\langle \int \mathcal{D}\vec{x} \prod_i p_i(x_i(0)) \delta(\text{equation of motion}) \right\rangle_{\vec{\eta}}, \quad (\text{A1})$$

where $\int \mathcal{D}\vec{x} = \prod_i \int \mathcal{D}x$ is a product of functional integrals, $\langle \dots \rangle_{\vec{\eta}} = (\prod_i \int \mathcal{D}\eta_i) \dots$ is the average over the noise realizations, and $\delta(\dots)$ is a functional Dirac delta function which restrict the integral to the paths that satisfy the stochastic dynamics. Using the Fourier representation of the delta function $\delta(x) \propto \int \mathcal{D}\hat{x} e^{-i \int dt \hat{x}(t)x(t)}$ we obtain

$$Z = \left\langle \int \mathcal{D}\vec{x} \mathcal{D}\vec{\hat{x}} \prod_i p_i(x_i(0)) e^{\int dt (-i\hat{x}_i(t)) [\dot{x}_i(t) - x_i(t)(1 - x_i + \sum_{j \neq i} a_{ij} J_{ij} x_j + h_i(t)) - \eta_i(t) - \lambda]} \right\rangle_{\vec{\eta}}. \quad (\text{A2})$$

The integral over time run over a fixed time interval, but we omit the limits of integration, here and elsewhere, for the sake of simplifying the notation. We introduce an auxiliary variable $f_i = 1 - x_i - \sum_{j \neq i} a_{ij} J_{ij} x_j(t) + h_i(t)$, which can be inserted in the calculation through a Dirac delta function. Integrating over the Gaussian noise we obtain

$$Z = \int \mathcal{D}\vec{x} \mathcal{D}\vec{\hat{x}} \mathcal{D}\vec{f} \mathcal{D}\vec{\hat{f}} \prod_{i \in V} p_i(x_i(0)) e^{S_i(x_i, \hat{x}_i, f_i, \hat{f}_i)} \prod_{(i,j) \in E} e^{S_{ij}(x_i, \hat{x}_i, f_i, \hat{f}_i, x_j, \hat{x}_j, f_j, \hat{f}_j)}, \quad (\text{A3})$$

where we have defined the local and interactive actions as

$$S_i = \int dt \left\{ -i\hat{x}_i(t) (\dot{x}_i(t) - x_i(t)f_i(t) - \lambda) - \frac{1}{2} 2Tg(x_i(t))\hat{x}_i^2(t) - i\hat{f}_i(t) [f_i(t) - (1 - x_i(t) + h_i(t))] \right\} \quad (\text{A4})$$

$$S_{ij} = - \int dt \left(J_{ij} i\hat{f}_i(t)x_j(t) + J_{ji} i\hat{f}_j(t)x_i(t) \right) \quad (\text{A5})$$

We put forward the following *dynamic cavity equations* as an ansatz for describing linearly coupled stochastic dynamics on sparse networks

$$c_{i \setminus j}[x_i, \hat{x}_i, f_i, \hat{f}_i] = \frac{1}{Z_{i \setminus j}} e^{S_i} \prod_{k \in \partial i \setminus j} \int \mathcal{D}x_k \mathcal{D}\hat{x}_k \mathcal{D}f_k \mathcal{D}\hat{f}_k c_{k \setminus i}[x_k, \hat{x}_k, f_k, \hat{f}_k] e^{S_{ki}}, \quad (\text{A6})$$

The full dynamic cavity marginals can be obtained from the cavity ones as follows

$$c_i[x_i, \hat{x}_i, f_i, \hat{f}_i] = \frac{1}{Z_i} e^{S_i} \prod_{k \in \partial i} \int \mathcal{D}x_k \mathcal{D}\hat{x}_k \mathcal{D}f_k \mathcal{D}\hat{f}_k c_{k \setminus i}[x_k, \hat{x}_k, f_k, \hat{f}_k] e^{S_{ki}}. \quad (\text{A7})$$

Expanding the cavity marginals Eq. (A6) at second order in the coupling parameters J_{ij} we obtain an effective single node action for the cavity marginal

$$c_{i \setminus j}[x_i, \hat{x}_i, f_i, \hat{f}_i] \approx \frac{p_i(x_i(0))}{Z_{i \setminus j}} e^{S_{i \setminus j}^{\text{eff}}}, \quad (\text{A8})$$

where the effective cavity action is defined as

$$S_{i \setminus j}^{\text{eff}} = \int dt \left\{ -i\hat{x}_i(t) (\dot{x}_i(t) - x_i(t)f_i(t) - \lambda) - \frac{1}{2} 2Tg(x_i(t))\hat{x}_i^2(t) - i\hat{f}_i(t) [f_i(t) - (1 - x_i(t) + h_i(t))] \right. \\ \left. + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}(t) - \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int dt' R_{k \setminus i}(t, t') x_i(t') \right\} + \int dt dt' i\hat{f}(t) \left(\sum_{k \in \partial i \setminus j} J_{ik}^2 C_{k \setminus i}(t, t') \right) i\hat{f}(t') \quad (\text{A9})$$

where $\mu_{k \setminus i}(t)$, $R_{k \setminus i}(t, t')$, and $C_{k \setminus i}(t, t')$ are respectively the cavity mean abundance, the cavity response function, and the cavity (connected) two-point correlation function. They are obtained self-consistently from the effective dynamics of the neighbors

$$\mu_{k \setminus i}(t) = \langle x_k(t) \rangle_{k \setminus i}, \quad R_{k \setminus i}(t, t') = \left\langle x_k(t) \dot{f}_k(t') \right\rangle_{k \setminus i}, \quad C(t, t') = \langle x_k(t) x_k(t') \rangle_{k \setminus i} - \mu_{k \setminus i}(t) \mu_{k \setminus i}(t'), \quad (\text{A10})$$

where $\langle \dots \rangle_{k \setminus i} = \int \mathcal{D}x_k \mathcal{D}\hat{x}_k \mathcal{D}f_k \mathcal{D}\hat{f}_k c_{k \setminus i}[x_k, \hat{x}_k, f_k, \hat{f}_k] \dots$ describes the local average over the cavity marginal.

The effective cavity action can be identified with a sort of local effective single-edge action for the variables on site i when the interaction term with site j is removed from the underlying graph.

Performing backward the steps of the path-integral representation, we obtain an effective stochastic dynamics in terms of the stochastic differential equation on the cavity graph

$$\begin{aligned} \frac{d}{dt} x_{i \setminus j}(t) = x_{i \setminus j}(t) & \left[1 - x_{i \setminus j}(t) + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}(t) + \sum_{k \in \partial i \setminus j} \int_0^t dt' J_{ik} J_{ki} R_{k \setminus i}(t, t') x_{i \setminus j}(t') \right. \\ & \left. + h_{i \setminus j}(t) + \zeta_{i \setminus j}(t) \right] + \sqrt{2Tg(x_{i \setminus j})} \xi_{i \setminus j}(t) + \lambda, \end{aligned} \quad (\text{A11})$$

where $\zeta_{i \setminus j}$ and $\xi_{i \setminus j}$ are Gaussian noises with vanishing mean and correlations

$$\langle \zeta_{i \setminus j}(t) \zeta_{i \setminus j}(t') \rangle_{i \setminus j} = \sum_{k \in \partial i \setminus j} J_{ik}^2 C_{k \setminus i}(t, t'), \quad \langle \xi_{i \setminus j}(t) \xi_{i \setminus j}(t') \rangle_{i \setminus j} = \delta(t - t'). \quad (\text{A12})$$

The terms $\mu_{k \setminus i}(t)$, $R_{k \setminus i}(t, t')$, and $C_{k \setminus i}(t, t')$ have to be considered as external parameters, obtained self-consistently as averages over the effective dynamics of the neighbors. In a similar way, an effective process for the original graph can be obtained as

$$\begin{aligned} \frac{d}{dt} x_i(t) = x_i(t) & \left[1 - x_i(t) + \sum_{k \in \partial i} J_{ik} \mu_{k \setminus i}(t) + \sum_{k \in \partial i} \int_0^t dt' J_{ik} J_{ki} R_{k \setminus i}(t, t') x_i(t') \right. \\ & \left. + h_i(t) + \zeta_i(t) \right] + \sqrt{2Tg(x_i(t))} \xi_i(t) + \lambda, \end{aligned} \quad (\text{A13})$$

with Gaussian noises with vanishing mean and correlations

$$\langle \zeta_i(t) \zeta_i(t') \rangle_i = \sum_{k \in \partial i} J_{ik}^2 C_{k \setminus i}(t, t'), \quad \langle \xi_i(t) \xi_i(t') \rangle_i = \delta(t - t'). \quad (\text{A14})$$

Appendix B: Single-path response function

The single-path response function $R_{i \setminus j}^{\text{sp}}(t, t') = \delta / \delta h_{i \setminus j}(t') x_{i \setminus j}(t)$ can be obtained by directly applying the functional derivative $\delta / \delta h_{i \setminus j}(t')$ to Eq. (4):

$$\begin{aligned} \frac{\partial}{\partial t} R_{i \setminus j}^{\text{sp}}(t, t') &= R_{i \setminus j}^{\text{sp}}(t, t') (1 - 2x_{i \setminus j}(t)) + R_{i \setminus j}^{\text{sp}}(t, t') \sqrt{\frac{2T}{g(x_{i \setminus j}(t))}} g'(x_{i \setminus j}(t)) \xi_{i \setminus j}(t) \\ &+ R_{i \setminus j}^{\text{sp}}(t, t') \left(\sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}(t) + \zeta_{i \setminus j}(t) \right) + x_{i \setminus j}(t) \delta(t - t') \\ &+ R_{i \setminus j}^{\text{sp}}(t, t') \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^t ds R_{k \setminus i}(t, s) x_{i \setminus j}(s) + x_{i \setminus j}(t) \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^t ds R_{k \setminus i}(t, s) R_{i \setminus j}^{\text{sp}}(s, t'). \end{aligned} \quad (\text{B1})$$

The single-path response function can therefore be integrated numerically using the initial condition $R_{i \setminus j}^{\text{sp}}(t, t') = 0$ for $t < t'$ from causality.

Appendix C: Numerical integration of GECaM dynamics

The numerical integration of the GECaM dynamics is performed using an Euler-Maruyama scheme with a time step Δ . Time is discretized in intervals of size Δ , i.e. $t = n\Delta$ with $n = 0, 1, \dots, T$, where T is the number of time steps. The trajectories $x(t)$ become T -dimensional time-vectors with elements $x^n = x(t = n\Delta)$, while the self-consistent fields $\mu(t)$, $R(t, t')$ and $C(t, t')$ become, respectively, T -dimensional vectors and $T \times T$ matrices, with elements $\mu^n = \mu(t = n\Delta)$, $R^{n,l} = R(t = n\Delta, t' = l\Delta)$ and $C^{n,l} = C(t = n\Delta, t' = l\Delta)$.

Both the effective cavity SDEs Eq. (4) and the effective marginal SDEs Eq. (7) are better integrated in log space. The discretized effective cavity SDE in log space reads

$$\log x_{i \setminus j}^{n+1} = \log x_{i \setminus j}^n + \Delta \mathcal{F}_{i \setminus j}^n \left(\mathbf{x}_{i \setminus j} \middle| \left\{ \mu_{k \setminus i}^n \right\}_{k \in \partial i \setminus j}, \left\{ \mathbf{R}_{k \setminus i} \right\}_{k \in \partial i \setminus j}, \zeta_{i \setminus j}^n \right) + \Delta \sqrt{2T} \mathcal{G} \left(x_{i \setminus j}^n \middle| \xi_{i \setminus j}^n \right) + \Delta \mathcal{H} \left(x_{i \setminus j}^n \middle| \lambda \right), \quad (\text{C1})$$

where

$$\mathcal{F}_{i \setminus j}^n \left(\mathbf{x}_{i \setminus j} \middle| \left\{ \mu_{k \setminus i}^n \right\}_{k \in \partial i \setminus j}, \left\{ \mathbf{R}_{k \setminus i} \right\}_{k \in \partial i \setminus j}, \zeta_{i \setminus j}^n \right) = 1 - x_{i \setminus j}^n + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}^n + \Delta \sum_{m=0}^{n-1} x_{i \setminus j}^m \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} R_{k \setminus i}^{n,m} + \zeta_{i \setminus j}^n, \quad (\text{C2})$$

$$\mathcal{G}(x | \xi) = \begin{cases} 0 & g(x) = 0 & \text{deterministic} \\ \exp[\log \xi - \frac{1}{2} \log x] & g(x) = x & \text{demographic} \\ \xi & g(x) = x^2 & \text{environmental}, \end{cases} \quad (\text{C3})$$

$$\mathcal{H}(x | \lambda) = \begin{cases} 0 & \lambda = 0 \\ \exp[\log \lambda - \log x] & \lambda > 0. \end{cases} \quad (\text{C4})$$

The discretized equation for the single path cavity response function reads

$$\begin{aligned} R_{i \setminus j}^{(\text{sp})n+1,l} &= R_{i \setminus j}^{(\text{sp})n,l} + \Delta R_{i \setminus j}^{(\text{sp})n,l} \left(1 - 2x_{i \setminus j}^n + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}^n + \zeta_{i \setminus j}^n + \sqrt{2T} \xi_{i \setminus j}^n \mathcal{I}(x_{i \setminus j}^n) \right) + x_{i \setminus j}^n \delta_{n,l} \\ &+ \Delta^2 R_{i \setminus j}^{(\text{sp})n,l} \sum_{m=0}^{n-1} x_{i \setminus j}^m \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} R_{k \setminus i}^{n,m} + \Delta^2 x_{i \setminus j}^n \sum_{m=l+1}^{n-1} R_{i \setminus j}^{(\text{sp})m,l} \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} R_{k \setminus i}^{n,m}, \end{aligned} \quad (\text{C5})$$

where

$$\mathcal{I}(x) = \begin{cases} 0 & g(x) = 0 & \text{deterministic} \\ 1/\sqrt{x} & g(x) = x & \text{demographic} \\ 1 & g(x) = x^2 & \text{environmental}. \end{cases} \quad (\text{C6})$$

The discretized equation for the single path marginal response function reads

$$\begin{aligned} R_i^{(\text{sp})n+1,l} &= R_i^{(\text{sp})n,l} + \Delta R_i^{(\text{sp})n,l} \left(1 - 2x_i^n + \sum_{k \in \partial i} J_{ik} \mu_{k \setminus i}^n + \zeta_i^n + \sqrt{2T} \xi_i^n \mathcal{I}(x_i^n) \right) + x_i^n \delta_{n,l} \\ &+ \Delta^2 R_i^{(\text{sp})n,l} \sum_{m=0}^{n-1} x_i^m \sum_{k \in \partial i} J_{ik} J_{ki} R_{k \setminus i}^{n,m} + \Delta^2 x_i^n \sum_{m=l+1}^{n-1} R_i^{(\text{sp})m,l} \sum_{k \in \partial i} J_{ik} J_{ki} R_{k \setminus i}^{n,m}. \end{aligned} \quad (\text{C7})$$

The discretized effective marginal SDE in log space reads

$$\log x_i^{n+1} = \log x_i^n + \Delta \mathcal{F}_i^n \left(\mathbf{x}_i \middle| \left\{ \mu_{k \setminus i}^n \right\}_{k \in \partial i}, \left\{ \mathbf{R}_{k \setminus i} \right\}_{k \in \partial i}, \zeta_i^n \right) + \Delta \sqrt{2T} \mathcal{G} \left(x_i^n \middle| \xi_i^n \right) + \Delta \mathcal{H} \left(x_i^n \middle| \lambda \right), \quad (\text{C8})$$

where

$$\mathcal{F}_i^n \left(\mathbf{x}_i \middle| \left\{ \mu_{k \setminus i}^n \right\}_{k \in \partial i}, \left\{ \mathbf{R}_{k \setminus i} \right\}_{k \in \partial i}, \zeta_i^n \right) = 1 - x_i^n + \sum_{k \in \partial i} J_{ik} \mu_{k \setminus i}^n + \Delta \sum_{m=0}^{n-1} x_i^m \sum_{k \in \partial i} J_{ik} J_{ki} R_{k \setminus i}^{n,m} + \zeta_i^n. \quad (\text{C9})$$

195 In order to avoid unnecessary computations, we can define the following node quantities:

$$\Sigma_{i,\mu}^n = \sum_{k \in \partial i} J_{ik} \mu_{k \setminus i}^n, \quad (\text{C10})$$

$$\Sigma_{i,R}^{n,l} = \sum_{k \in \partial i} J_{ik} J_{ki} R_{k \setminus i}^{n,l}, \quad (\text{C11})$$

$$\Sigma_{i,C}^{n,l} = \sum_{k \in \partial i} J_{ik}^2 C_{k \setminus i}^{n,l}, \quad (\text{C12})$$

$$(\text{C13})$$

196 which will be computed once at the beginning of the integration. The cavity contributions can therefore be computed
197 by subtracting the contribution of the node j from the sum over the neighbors of node i , thus avoiding the need to
198 compute the sum over all neighbors at each time step.

199 The resulting algorithm proceeds as follows.

200 1. Propose initial guesses for the cavity mean abundances $\{\mu_{i \setminus j}\}_{(i,j) \in E}$, the cavity correlations $\{C_{i \setminus j}\}_{(i,j) \in E}$, and
201 the cavity responses $\{R_{i \setminus j}\}_{(i,j) \in E}$ for each edge of the graph.

202 2. For each node i of the graph:

203 (a) Compute the node quantities $\Sigma_{i,\mu}^n$, $\Sigma_{i,R}^{n,l}$ and $\Sigma_{i,C}^{n,l}$ using Eqs. (C10), (C11) and (C12).

204 (b) For each edge (i, j) of the graph:

205 i. Sample N_{traj} Gaussian paths $\{(\zeta_{i \setminus j}^{(s)}, \xi_{i \setminus j}^{(s)})\}_{s=1}^{N_{\text{traj}}}$, where $\zeta_{i \setminus j}$ and $\xi_{i \setminus j}$ have, respectively, covariance
206 matrices $C_{i \setminus j}^{(\zeta)n,l} = \Sigma_{i,C}^{n,l} - J_{ik}^2 C_{k \setminus i}^{n,l}$ and $C_{i \setminus j}^{(\xi)n,l} = \delta_{n,l}$. This can be done by diagonalizing the covariance
207 matrix $C_{i \setminus j}^{(\zeta)}$ and using the proper basis to sample independent gaussian components.

208 ii. For each path $(\zeta_{i \setminus j}^{(s)}, \xi_{i \setminus j}^{(s)})$, numerically integrate the effective cavity equation with an initial condition
209 sampled according to $p_0(x)$ using the Euler-Maruyama scheme in log space, i.e. Eq. (C1). This will
210 yield N_{traj} trajectories $\{x_{i \setminus j}^{(s)}\}_{s=1}^{N_{\text{traj}}}$.

211 iii. For each path $(\zeta_{i \setminus j}^{(s)}, \xi_{i \setminus j}^{(s)})$, compute the single-path response function using Eq. (C5). This will yield
212 N_{traj} single path response matrices $\{R_{i \setminus j}^{(s)}\}_{s=1}^{N_{\text{traj}}}$.

213 iv. Compute the new values of the cavity self-consistent fields as averages over the paths:

$$\begin{aligned} \mu_{i \setminus j}^{(\text{new})n} &= \frac{1}{N_{\text{traj}}} \sum_{s=1}^{N_{\text{traj}}} x_{i \setminus j}^{(s)n}, \\ C_{i \setminus j}^{(\text{new})n,l} &= \frac{1}{N_{\text{traj}}} \sum_{s=1}^{N_{\text{traj}}} x_{i \setminus j}^{(s)n} x_{i \setminus j}^{(s)l} - \mu_{i \setminus j}^n \mu_{i \setminus j}^l, \\ R_{i \setminus j}^{(\text{new})n,l} &= \frac{1}{N_{\text{traj}}} \sum_{s=1}^{N_{\text{traj}}} R_{i \setminus j,k}^{(s)n,l}. \end{aligned}$$

214 3. Check for convergence of the cavity self-consistent fields.

$$\max_{(i,j)} \left\| C_{i \setminus j}^{\text{updated}} - C_{i \setminus j}^{\text{new}} \right\| < \epsilon, \quad (\text{C14})$$

215 If not converged, go to step 2.

216 4. For each node i of the graph:

217 (a) Compute the node quantities $\Sigma_{i,\mu}^n$, $\Sigma_{i,R}^{n,l}$ and $\Sigma_{i,C}^{n,l}$ using Eqs. (C10), (C11) and (C12).

218 (b) Sample N_{traj} Gaussian paths $\{(\zeta_i^{(s)}, \xi_i^{(s)})\}_{s=1}^{N_{\text{traj}}}$, where ζ_i and ξ_i have, respectively, covariance matrices
219 $C_i^{(\zeta)n,l} = \Sigma_{i,C}^{n,l}$ and $C_i^{(\xi)n,l} = \delta_{n,l}$. This can be done by diagonalizing the covariance matrix $C_i^{(\zeta)}$ and using
220 the proper basis to sample independent gaussian components.

- (c) For each path $(\zeta_i^{(s)}, \xi_i^{(s)})$, numerically integrate the effective marginal equation with an initial condition sampled according to $p_0(x)$ using the Euler-Maruyama scheme in log space, i.e. Eq. (C8). This will yield N_{traj} trajectories $\{x_i^{(s)}\}_{s=1}^{N_{\text{traj}}}$.
- (d) For each path $(\zeta_i^{(s)}, \xi_i^{(s)})$, compute the single-path response function using Eq. (C5). This will yield N_{traj} single path response matrices $\{R_i^{(s)}\}_{s=1}^{N_{\text{traj}}}$.
- (e) Compute the marginal self-consistent fields as averages over the paths:

$$\begin{aligned}\mu_i^n &= \frac{1}{N_{\text{traj}}} \sum_{s=1}^{N_{\text{traj}}} x_i^{(s)n}, \\ C_i^{n,l} &= \frac{1}{N_{\text{traj}}} \sum_{s=1}^{N_{\text{traj}}} x_i^{(s)n} x_i^{(s)l} - \mu_i^n \mu_i^l, \\ R_i^{n,l} &= \frac{1}{N_{\text{traj}}} \sum_{s=1}^{N_{\text{traj}}} R_{i,k}^{(s)n,l}.\end{aligned}$$

Appendix D: Gaussian closure

The effective process Eq. (7) is a stochastic differential equation that can be solved numerically. However, it is often useful to obtain analytical closed form equations for the fields μ , C and R . For simplicity, we rewrite the effective cavity process in the form

$$\frac{d}{dt} x_{i \setminus j}(t) = x_{i \setminus j}(t) \left[1 - x_{i \setminus j}(t) + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}(t) + \sum_{k \in \partial i \setminus j} \int_0^t dt' J_{ik} J_{ki} R_{k \setminus i}(t, t') x_{i \setminus j}(t') \right] + \eta_{i \setminus j}(t) + \lambda, \quad (\text{D1})$$

where we have defined the noise term $\eta_{i \setminus j}(t)$ as a Gaussian process with vanishing mean and covariance

$$\langle \eta_{i \setminus j}(t) \eta_{i \setminus j}(t') \rangle = x_{i \setminus j}(t) \sum_{k \in \partial i \setminus j} J_{ik}^2 C_{k \setminus i}(t, t') x_{i \setminus j}(t') + 2Tg(x_{i \setminus j}(t)) \delta(t - t'). \quad (\text{D2})$$

The approximated cavity message can be written as

$$\begin{aligned}c_{i \setminus j}[x_{i \setminus j}, \hat{x}_{i \setminus j}] &= \left\langle \exp \int dt \left\{ -i \hat{x}_{i \setminus j}(t) \left[\frac{\dot{x}_{i \setminus j}(t)}{x_{i \setminus j}(t)} - \left(1 - x_{i \setminus j}(t) + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}(t) \right. \right. \right. \right. \\ &\quad \left. \left. \left. + \sum_{k \in \partial i \setminus j} \int_0^t ds J_{ik} J_{ki} R_{k \setminus i}(t, s) x_{i \setminus j}(s) + \frac{\eta_{i \setminus j}(t)}{x_{i \setminus j}(t)} + \frac{\lambda}{x_{i \setminus j}(t)} \right) \right] \right\} \right\rangle\end{aligned} \quad (\text{D3a})$$

$$\begin{aligned}&= \exp \left\{ - \int dt \frac{i \hat{x}_{i \setminus j}(t)}{x_{i \setminus j}(t)} \left[\dot{x}_{i \setminus j}(t) - x_{i \setminus j}(t) \left(1 - x_{i \setminus j}(t) + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}(t) \right. \right. \right. \\ &\quad \left. \left. \left. + \sum_{k \in \partial i \setminus j} \int_0^t ds J_{ik} J_{ki} R_{k \setminus i}(t, s) x_{i \setminus j}(s) \right) - \lambda \right] \right\} \left\langle \exp \int dt \frac{i \hat{x}_{i \setminus j}(t)}{x_{i \setminus j}(t)} \eta_{i \setminus j}(t) \right\rangle\end{aligned} \quad (\text{D3b})$$

$$\begin{aligned}&= \exp \left\{ - \int dt \frac{i \hat{x}_{i \setminus j}(t)}{x_{i \setminus j}(t)} \left[\dot{x}_{i \setminus j}(t) - x_{i \setminus j}(t) \left(1 - x_{i \setminus j}(t) + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k \setminus i}(t) \right. \right. \right. \\ &\quad \left. \left. \left. + \sum_{k \in \partial i \setminus j} \int_0^t ds J_{ik} J_{ki} R_{k \setminus i}(t, s) x_{i \setminus j}(s) \right) - \lambda \right] \right\} \\ &\quad + \frac{1}{2} \int dt dt' \frac{i \hat{x}_{i \setminus j}(t)}{x_{i \setminus j}(t)} \left(2Tg(x_{i \setminus j}(t)) \delta(t - t') + x_{i \setminus j}(t) \sum_{k \in \partial i \setminus j} J_{ik}^2 C_{k \setminus i}(t, t') x_{i \setminus j}(t') \right) \frac{i \hat{x}_{i \setminus j}(t')}{x_{i \setminus j}(t')} \Bigg\}. \quad (\text{D3c})\end{aligned}$$

233 By averaging Eq. (4) over the noise realizations, we obtain the following self-consistent equations for the cavity
 234 mean abundances

$$\begin{aligned} \frac{d}{dt}\mu_{i\setminus j}(t) &= \langle \dot{x}_{i\setminus j}(t) \rangle = \mu_{i\setminus j}(t) - C_{i\setminus j}(t, t) - \mu_{i\setminus j}^2(t) + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k\setminus i}(t) \mu_{i\setminus j}(t) \\ &+ \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^t ds R_{k\setminus i}(t, s) (C_{i\setminus j}(t, s) + \mu_{i\setminus j}(t) \mu_{i\setminus j}(s)) + \langle \eta_{i\setminus j}(t) \rangle + \lambda \end{aligned} \quad (D4a)$$

$$\begin{aligned} &= \mu_{i\setminus j}(t) - C_{i\setminus j}(t, t) - \mu_{i\setminus j}^2(t) + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k\setminus i}(t) \mu_{i\setminus j}(t) \\ &+ \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^t ds R_{k\setminus i}(t, s) (C_{i\setminus j}(t, s) + \mu_{i\setminus j}(t) \mu_{i\setminus j}(s)) + \lambda. \end{aligned} \quad (D4b)$$

235 Similarly, we can obtain the self-consistent equations for the disconnected cavity correlations

$$\begin{aligned} \frac{\partial}{\partial t} C_{i\setminus j}^{\text{dc}}(t, t') &= \langle \dot{x}_{i\setminus j}(t) x_{i\setminus j}(t') \rangle = C_{i\setminus j}^{\text{dc}}(t, t') - \langle x_{i\setminus j}^2(t) x_{i\setminus j}(t') \rangle + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k\setminus i}(t) C_{i\setminus j}^{\text{dc}}(t, t') \\ &+ \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^t ds R_{k\setminus i}(t, s) \langle x_{i\setminus j}(t) x_{i\setminus j}(s) x_{i\setminus j}(t') \rangle + \langle \eta_{i\setminus j}(t) x_{i\setminus j}(t') \rangle + \lambda \mu_{i\setminus j}(t') \end{aligned} \quad (D5a)$$

$$\begin{aligned} &= C_{i\setminus j}^{\text{dc}}(t, t') - \langle x_{i\setminus j}^2(t) x_{i\setminus j}(t') \rangle + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k\setminus i}(t) C_{i\setminus j}^{\text{dc}}(t, t') \\ &+ \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int_0^t ds R_{k\setminus i}(t, s) \langle x_{i\setminus j}(t) x_{i\setminus j}(s) x_{i\setminus j}(t') \rangle + 2T \langle \frac{g(x_{i\setminus j}(t))}{x_{i\setminus j}(t)} \dot{x}_{i\setminus j}(t) x_{i\setminus j}(t') \rangle \\ &+ \sum_{k \in \partial i \setminus j} J_{ik}^2 \int ds C_{k\setminus i}(t, s) \langle x_{i\setminus j}(t) \dot{x}_{i\setminus j}(s) x_{i\setminus j}(t') \rangle + \lambda \mu_{i\setminus j}(t'). \end{aligned} \quad (D5b)$$

236 The self-consistent equations for the cavity response functions can be obtained as

$$\begin{aligned} \frac{\partial}{\partial t} R_{i\setminus j}(t, t') &= \langle \dot{x}_{i\setminus j}(t) \dot{x}_{i\setminus j}(t') \rangle = R_{i\setminus j}(t, t') - \langle x_{i\setminus j}^2(t) \dot{x}_{i\setminus j}(t') \rangle + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k\setminus i}(t) R_{i\setminus j}(t, t') \\ &+ \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int ds R_{k\setminus i}(t, s) \langle x_{i\setminus j}(t) x_{i\setminus j}(s) \dot{x}_{i\setminus j}(t') \rangle + \langle \eta_{i\setminus j}(t) \dot{x}_{i\setminus j}(t') \rangle. \end{aligned} \quad (D6a)$$

$$\begin{aligned} &= R_{i\setminus j}(t, t') - \langle x_{i\setminus j}^2(t) \dot{x}_{i\setminus j}(t') \rangle + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k\setminus i}(t) R_{i\setminus j}(t, t') \\ &+ \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \int ds R_{k\setminus i}(t, s) \langle x_{i\setminus j}(t) x_{i\setminus j}(s) \dot{x}_{i\setminus j}(t') \rangle + 2T \langle \frac{g(x_{i\setminus j}(t))}{x_{i\setminus j}(t)} \dot{x}_{i\setminus j}(t) \dot{x}_{i\setminus j}(t') \rangle \\ &+ \sum_{k \in \partial i \setminus j} J_{ik}^2 \int ds C_{k\setminus i}(t, s) \langle x_{i\setminus j}(t) \dot{x}_{i\setminus j}(s) \dot{x}_{i\setminus j}(t') \rangle. \end{aligned} \quad (D6b)$$

237 The equations for the correlations and responses are not closed, as they depend on three-, four- and five-times averages.

238 We can close the equations by approximating those averages with sums of products of two-times averages, i.e. we can

239 use the Gaussian closure approximation. Specifically, introducing the zero-mean degrees of freedom $\delta x(t) = x(t) - \mu(t)$,

240 we can approximate as:

$$\begin{aligned} \langle x^2(t) x(t') \rangle &= \langle (\delta x(t) + \mu(t))^2 (\delta x(t') + \mu(t')) \rangle \approx C(t, t) \mu(t') + \mu^2(t) \mu(t') + 2\mu(t) C(t, t'), \\ \langle x(t) x(s) x(t') \rangle &= \langle (\delta x(t) + \mu(t)) (\delta x(s) + \mu(s)) (\delta x(t') + \mu(t')) \rangle \\ &\approx C(t, s) \mu(t') + C(t, t') \mu(s) + C(s, t') \mu(t) + \mu(t) \mu(s) \mu(t'), \\ \langle x(t) \dot{x}(s) x(t') \rangle &= \langle (\delta x(t) + \mu(t)) \dot{x}(s) (\delta x(t') + \mu(t')) \rangle \approx \mu(t) R(t', s) + R(t, s) \mu(t'), \\ \langle x^2(t) \dot{x}(t') \rangle &= \langle (\delta x(t) + \mu(t))^2 \dot{x}(t') \rangle \approx 2\mu(t) R(t, t'), \\ \langle x(t) x(s) \dot{x}(t') \rangle &= \langle (\delta x(t) + \mu(t)) (\delta x(s) + \mu(s)) \dot{x}(t') \rangle \approx R(t, t') \mu(s) + \mu(t) R(s, t'), \\ \langle x(t) \dot{x}(s) \dot{x}(t') \rangle &= \langle (\delta x(t) + \mu(t)) \dot{x}(s) \dot{x}(t') \rangle \approx 0. \end{aligned}$$

241 The terms involving the function $g(x)$ depend on the specific form of the function.

242 • **Demographic:** $g(x) = x$

$$\begin{aligned}\langle \frac{g(x(t))}{x(t)} \hat{i}x(t)x(t') \rangle &= \langle \hat{i}x(t)x(t') \rangle = R(t', t), \\ \langle \frac{g(x(t))}{x(t)} \hat{i}x(t)\hat{i}x(t') \rangle &= \langle \hat{i}x(t)\hat{i}x(t') \rangle = 0.\end{aligned}$$

243 • **Environmental:** $g(x) = x^2$

$$\begin{aligned}\langle \frac{g(x(t))}{x(t)} \hat{i}x(t)x(t') \rangle &= \langle (\delta x(t) + \mu(t))\hat{i}x(t)(\delta x(t') + \mu(t')) \rangle \approx \mu(t)R(t', t), \\ \langle \frac{g(x(t))}{x(t)} \hat{i}x(t)\hat{i}x(t') \rangle &= \langle (\delta x(t) + \mu(t))\hat{i}x(t)\hat{i}x(t') \rangle \approx 0.\end{aligned}$$

244 The resulting closed equations for the cavity mean abundances, correlations and responses read

$$\begin{aligned}\frac{d}{dt}\mu_{i\setminus j}(t) &= \mu_{i\setminus j}(t) - C_{i\setminus j}(t, t) - \mu_{i\setminus j}^2(t) + \sum_{k \in \partial i \setminus j} J_{ik}\mu_{k\setminus i}(t)\mu_{i\setminus j}(t) \\ &+ \sum_{k \in \partial i \setminus j} J_{ik}J_{ki} \int_0^t ds R_{k\setminus i}(t, s) (C_{i\setminus j}(t, s) + \mu_{i\setminus j}(t)\mu_{i\setminus j}(s)) + \lambda,\end{aligned}\quad (D7)$$

$$\begin{aligned}\frac{\partial}{\partial t}C_{i\setminus j}^{\text{dc}}(t, t') &= C_{i\setminus j}^{\text{dc}}(t, t') (1 - 2\mu_{i\setminus j}(t)) - C_{i\setminus j}^{\text{dc}}(t, t)\mu_{i\setminus j}(t') + 2\mu_{i\setminus j}^2(t)\mu_{i\setminus j}(t') + \sum_{k \in \partial i \setminus j} J_{ik}\mu_{k\setminus i}(t)C_{i\setminus j}^{\text{dc}}(t, t') \\ &+ \sum_{k \in \partial i \setminus j} J_{ik}J_{ki} \int_0^t ds R_{k\setminus i}(t, s) (C_{i\setminus j}^{\text{dc}}(t, s)\mu_{i\setminus j}(t') + \mu_{i\setminus j}(t)C_{i\setminus j}^{\text{dc}}(s, t') + C_{i\setminus j}^{\text{dc}}(t, t')\mu_{i\setminus j}(s)) \\ &- 2 \sum_{k \in \partial i \setminus j} J_{ik}J_{ki} \int_0^t ds R_{k\setminus i}(t, s)\mu_{i\setminus j}(t)\mu_{i\setminus j}(s)\mu_{i\setminus j}(t') + 2T\mathcal{L}_{i\setminus j}(t)R_{i\setminus j}(t', t) \\ &+ \sum_{k \in \partial i \setminus j} J_{ik}^2 \int ds C_{k\setminus i}(t, s) (\mu_{i\setminus j}(t)R_{i\setminus j}(t', s) + R_{i\setminus j}(t, s)\mu_{i\setminus j}(t')) + \lambda\mu_{i\setminus j}(t').\end{aligned}\quad (D8)$$

$$\begin{aligned}\frac{\partial}{\partial t}R_{i\setminus j}(t, t') &= R_{i\setminus j}(t, t') (1 - 2\mu_{i\setminus j}(t)) + \sum_{k \in \partial i \setminus j} J_{ik}\mu_{k\setminus i}(t)R_{i\setminus j}(t, t') \\ &+ \sum_{k \in \partial i \setminus j} J_{ik}J_{ki} \int ds R_{k\setminus i}(t, s) (R_{i\setminus j}(t, t')\mu_{i\setminus j}(s) + \mu_{i\setminus j}(t)R_{i\setminus j}(s, t')).\end{aligned}\quad (D9)$$

245 where we have introduced the function

$$\mathcal{L}_{i\setminus j}(t) = \begin{cases} 0 & g(x) = 0 & \text{deterministic} \\ 1 & g(x) = x & \text{demographic} \\ \mu_{i\setminus j}(t) & g(x) = x^2 & \text{environmental.} \end{cases}\quad (D10)$$

246 We can finally obtain the self-consistent equation for the connected cavity correlations by subtracting $\dot{\mu}_{i\setminus j}(t)\mu_{i\setminus j}(t)$
247 from the self-consistent equation for the disconnected cavity correlations:

$$\begin{aligned}\frac{\partial}{\partial t}C_{i\setminus j}(t, t') &= (1 - 2\mu_{i\setminus j}(t)) C_{i\setminus j}(t, t') + \sum_{k \in \partial i \setminus j} J_{ik}\mu_{k\setminus i}(t)C_{i\setminus j}(t, t') \\ &+ \sum_{k \in \partial i \setminus j} J_{ik}J_{ki} \int_0^t ds R_{k\setminus i}(t, s) (\mu_{i\setminus j}(t)C_{i\setminus j}(s, t') + C_{i\setminus j}(t, t')\mu_{i\setminus j}(s)) \\ &+ 2T\mathcal{L}_{i\setminus j}(t)R_{i\setminus j}(t', t) + \sum_{k \in \partial i \setminus j} J_{ik}^2 \int_0^{\max(t, t')} ds C_{k\setminus i}(t, s) (\mu_{i\setminus j}(t)R_{i\setminus j}(t', s) + R_{i\setminus j}(t, s)\mu_{i\setminus j}(t')).\end{aligned}\quad (D11)$$

249 When the average over disordered interactions is taken into account, one has to deal with an integral of the form

$$\overline{\int \mathcal{D}y \mathcal{D}\hat{y} \mathcal{D}w \mathcal{D}\hat{w} c[y, \hat{y}, w, \hat{w}] \exp \left[-i \int dt \left(J\hat{w}(t)x(t) + J'\hat{f}(t)y(t) \right) \right]}, \quad (\text{E1})$$

250 where $\overline{\cdots}$ denotes the average over disordered couplings (J, J') , i.e. $\overline{\cdots} = \int dJ dJ' p(J, J') \cdots$, where $p(J, J')$ is a
 251 bivariate Gaussian probability distribution function with averages $\overline{J} = \overline{J'} = m/K$, variances $\overline{(J - \overline{J})^2} = \overline{(J' - \overline{J'})^2} =$
 252 σ^2/K and correlation $\overline{(J - \overline{J})(J' - \overline{J'})} = \gamma\sigma^2/K$.

253 Here we demonstrate that when dealing with a large connectivity expansion the average over disorder can be
 254 exchanged with the small-coupling expansion, provided one only retains terms of order $1/K$ in the expansion.

255 We use the following shorthand notation for the sake of shortness: $\mathcal{D}c = \mathcal{D}y \mathcal{D}\hat{y} \mathcal{D}w \mathcal{D}\hat{w} c[y, \hat{y}, w, \hat{w}]$, $a \cdot b =$
 256 $\int dt a(t)b(t)$, with a and b being $x, y, \hat{f}$ and $\hat{w}$.

257 **Average over disorder first**

258 Since the average over disorder involves a Gaussian integral, it can be easily computed as follows

$$\overline{\int \mathcal{D}c e^{-i(J\hat{w} \cdot x + J'\hat{f} \cdot y)}} = \int \mathcal{D}c e^{-i(J\hat{w} \cdot x + J'\hat{f} \cdot y)} \quad (\text{E2})$$

$$= \int \mathcal{D}c \exp \left[-\frac{m}{K} (i\hat{w} \cdot x + i\hat{f} \cdot y) - \frac{\sigma^2}{2K} \left((\hat{w} \cdot x)^2 + (\hat{f} \cdot y)^2 - 2\gamma(i\hat{w} \cdot x)(i\hat{f} \cdot y) \right) \right]. \quad (\text{E3})$$

259 Expanding at first order in $1/K$, i.e. large connectivity expansion

$$\overline{\int \mathcal{D}c e^{-i(J\hat{w} \cdot x + J'\hat{f} \cdot y)}} = \int \mathcal{D}c \left[1 - \frac{m}{K} (i\hat{w} \cdot x + i\hat{f} \cdot y) - \frac{\sigma^2}{2K} \left((\hat{w} \cdot x)^2 + (\hat{f} \cdot y)^2 - 2\gamma(i\hat{w} \cdot x)(i\hat{f} \cdot y) \right) \right] \quad (\text{E4})$$

$$\approx 1 - \frac{m}{K} \int dt \left(\langle i\hat{w}(t) \rangle x(t) + i\hat{f}(t) \langle y(t) \rangle \right) + \gamma \frac{\sigma^2}{K} \int dt dt' \langle y(t) i\hat{w}(t') \rangle i\hat{f}(t) x(t') \\ - \frac{\sigma^2}{2K} \int dt dt' \left(\langle \hat{w}(t) \hat{w}(t') \rangle x(t) x(t') + \hat{f}(t) \hat{f}(t') \langle y(t) y(t') \rangle \right) \quad (\text{E5})$$

$$= 1 - \frac{m}{K} \int dt i\hat{f}(t) \mu(t) + \gamma \frac{\sigma^2}{K} \int dt dt' i\hat{f}(t) R(t, t') x(t') - \frac{\sigma^2}{2K} \int dt dt' \hat{f}(t) C^{\text{dc}}(t, t') \hat{f}(t'), \quad (\text{E6})$$

260 where we have defined the average over the measure $c[y, \hat{y}, w, \hat{w}]$ as $\langle \cdots \rangle$. We have introduced the average $\mu(t) = \langle y(t) \rangle$,
 261 the (disconnected) correlation function $C^{\text{dc}}(t, t') = \langle y(t) y(t') \rangle$ and the response function $R(t, t') = \langle y(t) i\hat{w}(t') \rangle$. The
 262 averages of hatted quantities only vanish due to causality constraints. By re-exponentiating we get the following
 263 approximate expression

$$\overline{\int \mathcal{D}c e^{-i(J\hat{w} \cdot x + J'\hat{f} \cdot y)}} \approx \exp \left\{ -\frac{m}{K} \int dt i\hat{f}(t) \mu(t) + \gamma \frac{\sigma^2}{K} \int dt dt' i\hat{f}(t) R(t, t') x(t') \right. \\ \left. - \frac{\sigma^2}{2K} \int dt dt' \hat{f}(t) C^{\text{dc}}(t, t') \hat{f}(t') \right\}. \quad (\text{E7})$$

265 We expand the exponential at second order in J , i.e. small coupling expansion,

$$\overline{\int \mathcal{D}c e^{-i(J\hat{w}\cdot x + J'\hat{f}\cdot y)}} \approx \overline{\int \mathcal{D}c \left[1 - \left(J\hat{w}\cdot x + J'\hat{f}\cdot y \right) + \frac{1}{2} \left(J\hat{w}\cdot x + J'\hat{f}\cdot y \right)^2 \right]} \quad (\text{E8})$$

$$= 1 - \int dt \left(J\langle \hat{w}(t) \rangle x(t) + J'\langle \hat{f}(t) \rangle y(t) \right) + \frac{1}{2} \int dt dt' \left(J^2 \langle \hat{w}(t) \hat{w}(t') \rangle x(t) x(t') \right. \\ \left. + J'^2 \langle \hat{f}(t) \hat{f}(t') \rangle y(t) y(t') + 2JJ' \langle y(t) \hat{w}(t') \rangle \langle \hat{f}(t) x(t') \rangle \right) \quad (\text{E9})$$

$$= 1 - \int dt J'\langle \hat{f}(t) \rangle \mu(t) + \frac{1}{2} \int dt dt' \left(J'^2 \langle \hat{f}(t) C^{\text{dc}}(t, t') \rangle \langle \hat{f}(t') \rangle + 2JJ' \langle \hat{f}(t) R(t, t') x(t') \rangle \right). \quad (\text{E10})$$

266 By re-exponentiating, we have to carefully consider the square of first order terms, i.e. $1 - \langle x \rangle + \langle x^2 \rangle / 2 \approx \exp(\langle x \rangle +$
267 $\langle x^2 \rangle / 2 - \langle x \rangle^2 / 2)$ when $x \approx 0$.

$$\overline{\int \mathcal{D}c e^{-i(J\hat{w}\cdot x + J'\hat{f}\cdot y)}} \approx \overline{\exp \left\{ - \int dt J'\langle \hat{f}(t) \rangle \mu(t) + \frac{1}{2} \int dt dt' J'^2 \langle \hat{f}(t) C(t, t') \rangle \langle \hat{f}(t') \rangle \right.} \\ \left. + \int dt dt' JJ' \langle \hat{f}(t) R(t, t') x(t') \rangle \right\}}, \quad (\text{E11})$$

268 where we have introduced the (connected) correlation function $C(t, t') = C^{\text{dc}}(t, t') - \mu(t)\mu(t')$. We defined, for the
269 sake of shortness, the terms $A = \int dt \langle \hat{f}(t) \rangle \mu(t)$, $B = \int dt dt' \langle \hat{f}(t) C(t, t') \rangle \langle \hat{f}(t') \rangle$, and $C = \int dt dt' \langle \hat{f}(t) R(t, t') x(t') \rangle$. By
270 Gaussian integration, we obtain

$$\overline{\int \mathcal{D}c e^{-i(J\hat{w}\cdot x + J'\hat{f}\cdot y)}} \approx \exp \left(-AJ' + \frac{B}{2}J'^2 + CJJ' \right) \quad (\text{E12})$$

$$= \frac{\exp \left[\frac{\sigma^2 (A^2 K^2 + 2AC(\gamma-1)Km - 2C^2(\gamma-1)m^2) + Km(m(B+2C) - 2AK)}{2K(-K\sigma^2(B+2C\gamma) + C^2(\gamma^2-1)\sigma^4 + K^2)} \right]}{\sqrt{1 + \frac{C^2(\gamma^2-1)\sigma^4 - K\sigma^2(B+2C\gamma)}{K^2}}}. \quad (\text{E13})$$

271 Retaining only the first order terms in $1/K$, i.e. large connectivity expansion

$$\overline{\int \mathcal{D}c e^{-i(J\hat{w}\cdot x + J'\hat{f}\cdot y)}} \approx 1 - \frac{m}{K}A + \frac{\sigma^2}{2K} (A^2 + B + 2\gamma C) \quad (\text{E14})$$

$$\approx \exp \left[-\frac{m}{K}A + \frac{\sigma^2}{2K} (A^2 + B + 2\gamma C) \right] \quad (\text{E15})$$

$$= \exp \left\{ -\frac{m}{K} \int dt \langle \hat{f}(t) \rangle \mu(t) + \frac{\sigma^2}{2K} \int dt dt' \langle \hat{f}(t) \rangle (\mu(t)\mu(t') + C(t, t')) \langle \hat{f}(t') \rangle \right. \\ \left. + \gamma \frac{\sigma^2}{K} \int dt dt' \langle \hat{f}(t) R(t, t') x(t') \rangle \right\} \quad (\text{E16})$$

$$= \exp \left\{ -\frac{m}{K} \int dt \langle \hat{f}(t) \rangle \mu(t) + \frac{\sigma^2}{2K} \int dt dt' \langle \hat{f}(t) C^{\text{dc}}(t, t') \rangle \langle \hat{f}(t') \rangle \right. \\ \left. + \gamma \frac{\sigma^2}{K} \int dt dt' \langle \hat{f}(t) R(t, t') x(t') \rangle \right\}, \quad (\text{E17})$$

272 which is the same approximate result obtained by taking first the average over disorder. We note that the same result
 273 is obtained if we take the average over disorder before re-exponentiating, i.e in Eq. (E10),

$$\overline{\int \mathcal{D}c e^{-i(J\hat{w}\cdot x + J'\hat{f}\cdot y)}} \approx 1 - \int dt \overline{J} i\hat{f}(t)\mu(t) + \frac{1}{2} \int dt dt' \left(\overline{J'^2} i\hat{f}(t)C^{\text{dc}}(t, t')i\hat{f}(t') + 2\overline{J}\overline{J'} i\hat{f}(t)R(t, t')x(t') \right) \quad (\text{E18})$$

$$\approx 1 - \int dt \frac{m}{K} i\hat{f}(t)\mu(t) + \frac{1}{2} \int dt dt' \left(\frac{\sigma^2}{K} i\hat{f}(t)C^{\text{dc}}(t, t')i\hat{f}(t') + 2\gamma \frac{\sigma^2}{K} i\hat{f}(t)R(t, t')x(t') \right) \quad (\text{E19})$$

$$\approx \exp \left\{ -\frac{m}{K} \int dt i\hat{f}(t)\mu(t) + \frac{\sigma^2}{2K} \int dt dt' i\hat{f}(t)C^{\text{dc}}(t, t')i\hat{f}(t') \right. \\ \left. + \gamma \frac{\sigma^2}{K} \int dt dt' i\hat{f}(t)R(t, t')x(t') \right\}. \quad (\text{E20})$$